

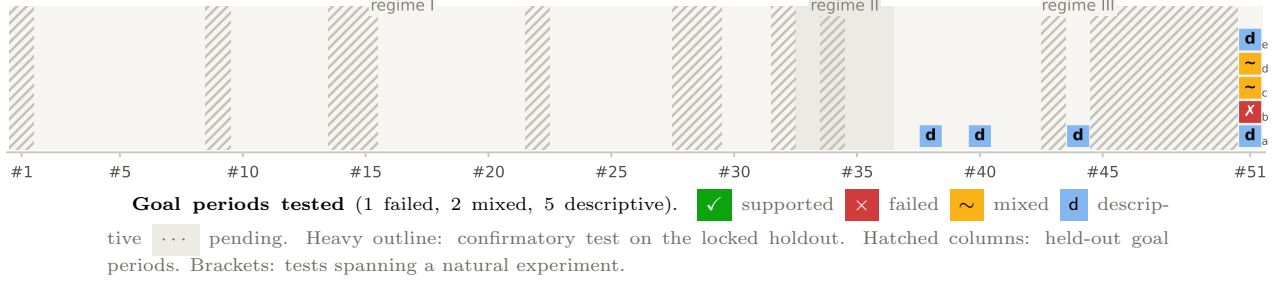
H22: Private, conflicting goals make #51 a spin glass

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

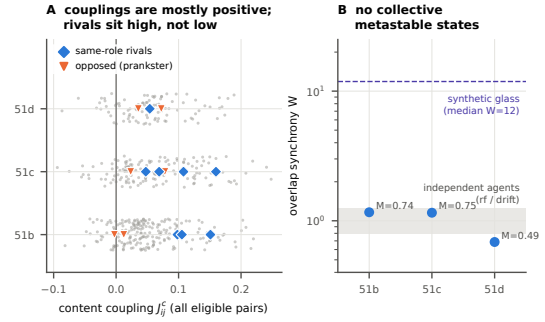
Bottom line. #51 is not a spin glass: private roles make it a random-field system with weak, mostly positive coupling, where rivals align rather than repel and the swarm never switches collectively.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Do #51's private, partly conflicting roles make the swarm a spin glass? Tested on sub-periods 51a-e (45 non-holdout days) against shared-objective weeks #38, #40, #44, after synthetic validation (the balance index separates factions from random signs; SK and a heterogeneous ferromagnet cannot be told apart under common drive). Content couplings are real in 51b and 51c ($\rho_{\text{split}} = 0.46$, $p = 0.003$) but 85–94% positive ($\hat{\kappa} = 4\text{--}10$); the balance index is indeterminate. Same-role rivals co-move *more* than unrelated pairs ($T_{\text{SR}} = +0.048$, $p = 0.02$), the homophily rival. Positions persist ($M = 0.5\text{--}0.75$) without collective switching ($W \approx 1$). Contrast weeks are no less ferromagnetic; talk couplings sit at the noise floor. Holdout test (#51 tail) written, **not run**.



Observables. (a) Every eligible pair's content coupling in the counted #51 units (gray), with same-role rivals (blue) and Prankster vs Ethicist/Psychologist pairs (orange): mostly positive, rivals high. (b) Overlap synchrony W at the independent-agent level (gray band), far below a synthetic glass (dashed). High memory M without synchrony: agents hold their own positions; the swarm never jumps between collective states.

PHYSICAL MODEL

Models 01 (SK) and 11 ($O(n)$ spins). Spin $\mathbf{s}_i$: agent i 's whitened chat embedding per 30-min window ($n = 32$). Fields: day field $\mathbf{m}_d$; static field $\mathbf{H}_i$ (role, lab, style). J_{ij} : correlation of within-day fluctuations $\mathbf{x}_i, \mathbf{x}_j$ minus a cross-day surrogate; A, B, C : day folds. κ : SK ratio J_0/J ; τ_3 : gauge-invariant balance (SK ≈ 0 , factions ≈ 0.5); q : overlap of day-demeaned states $\hat{\delta}$; W : its synchrony beyond per-agent shifts.

$$P(\{\mathbf{s}\}) \propto \exp \left(\sum_{i < j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (\mathbf{m}_d + \mathbf{H}_i) \cdot \mathbf{s}_i \right)$$

$$J_{ij} = \text{corr}_w(\mathbf{x}_i, \mathbf{x}_j) - \text{corr}_w^{\text{surr}}(\mathbf{x}_i, \mathbf{x}_j)$$

$$\kappa = \sqrt{N} \bar{J} / \sigma_J, \quad \tau_3 = \text{tr}(J^A J^B J^C) / \text{tr}(J^A J^B)^{3/2}$$

$$q(d, d') = \frac{1}{N} \sum_i \hat{\delta}_{i,d} \cdot \hat{\delta}_{i,d'}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Practically, little. “Spin glass” was the exciting reading of mixed-motive swarms: frustrated, history-dependent, hard to steer. The data say something duller and more usable. Assigned roles act as fixed fields that pin each agent's topic, the shared room adds a weak positive pull, and same-objective agents drift together. For an operator, roles, not inter-agent conflict, set what such a swarm talks about, and there are no hidden collective states to escape. Caveat: embeddings measure topic, not stance, so strategic opposition could be invisible here. The diagnostics port to other logs; the spin-glass vocabulary is not earned.

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RESULTS VS PREDICTIONS

Pending.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

Pending.

ROUND 2

GOAL PERIODS IN DETAIL

G38 d descriptive
G40 d descriptive
G44 d descriptive
G51a d descriptive
G51b × failed
G51c ~ inconclusive
G51d ~ inconclusive
G51e d descriptive

- **Essence.** Do conflicting incentives produce conflict behavior at all?
- **H22-R1** Cooperation default: agents help rivals at rates comparable to aligned pairs (Jev-labeled help acts).
- **H22-R2** A quantitative random-field fit (HH129): roles as fields plus weak positive coupling.

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